

O&G · 4 — History matching a well test: recovering permeability and skin ⇌

A pressure drawdown test is one of the oldest inverse problems in petroleum engineering: flow a well at a constant rate, record the bottomhole pressure, and back out the reservoir permeability k and the near-wellbore skin factor s .

For a well producing at constant rate in an infinite-acting reservoir, the semi-log ("MDH") approximation to the radial diffusivity equation's line-source solution is, in standard oilfield units (q in STB/d, B in rb/STB, μ in cp, k in md, h in ft, t in hours, ϕ a fraction, c_t in psi^{-1} , r_w in ft, p in psi):

$$p_{wf}(t) = p_i - 162.6 \frac{qB\mu}{kh} \left[\log_{10}(t) + \log_{10} \left(\frac{k}{\phi\mu c_t r_w^2} \right) - 3.23 + 0.87 s \right]$$

Everything except k and s is either measured or known from core/PVT data — this notebook estimates k and s from a synthetic noisy buildup, the same way an engineer reads a type curve or fits a straight line.

```
1 using Optimization, OptimizationOptimJL, ForwardDiff, Plots, Random, PlutoUI
```

1. The forward model ⇌

pwf_model (generic function with 1 method)

```
1 function pwf_model(t, k, s; pi_res=4200.0, q=400.0, B=1.2, mu=1.5, h=60.0, phi=0.18,
2   ct=1.2e-5, rw=0.3)
3   Δp = 162.6 * (q * B * mu) / (k * h) * (log10(t) + log10(k / (phi * mu * ct * rw^2)) -
4     3.23 + 0.87 * s)
5   return pi_res - Δp
6 end
```

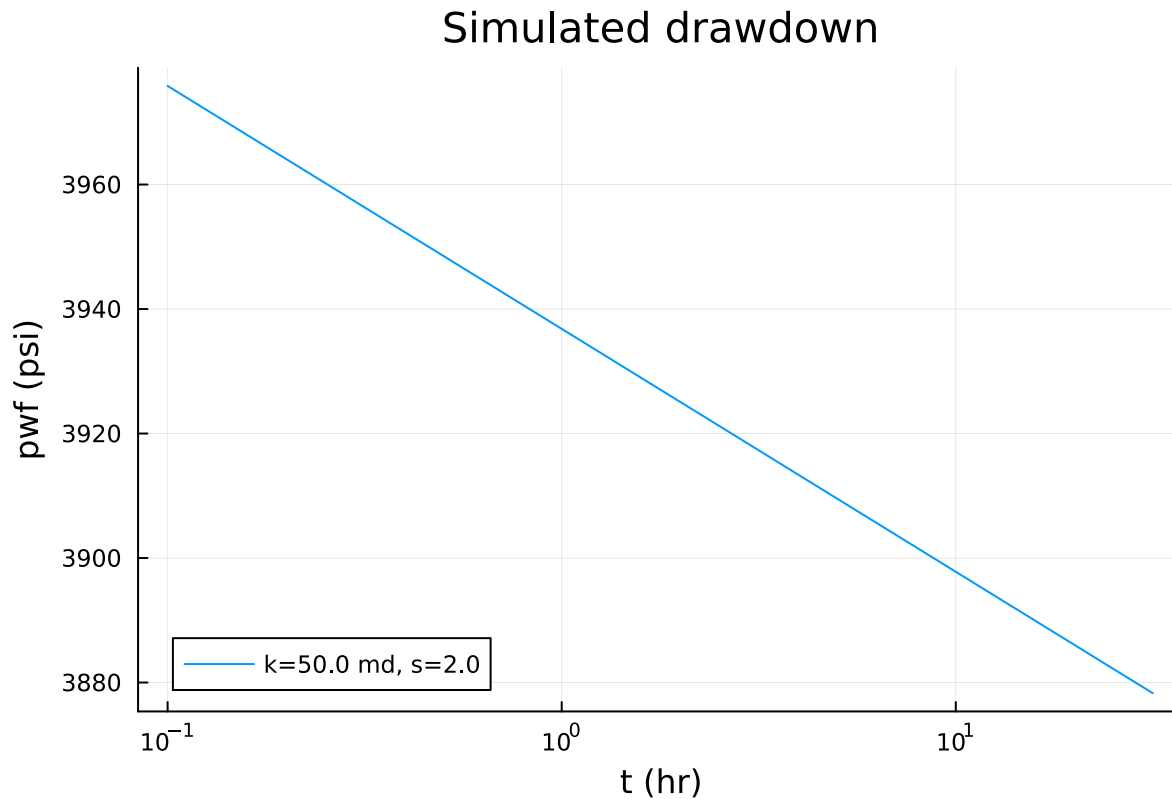
2. Play with the model ⇌

Skin controls the *extra* pressure drop near the wellbore (damage if $s > 0$, stimulation if $s < 0$); permeability controls how fast the transient pressure wave dissipates. Drag the sliders and watch the drawdown curve respond.

 50.0

```
1 @bind k_demo Slider(1.0:1.0:200.0, default=50.0, show_value=true)
```

```
1 @bind s_demo Slider(-3.0:0.5:15.0, default=2.0, show_value=true)
```



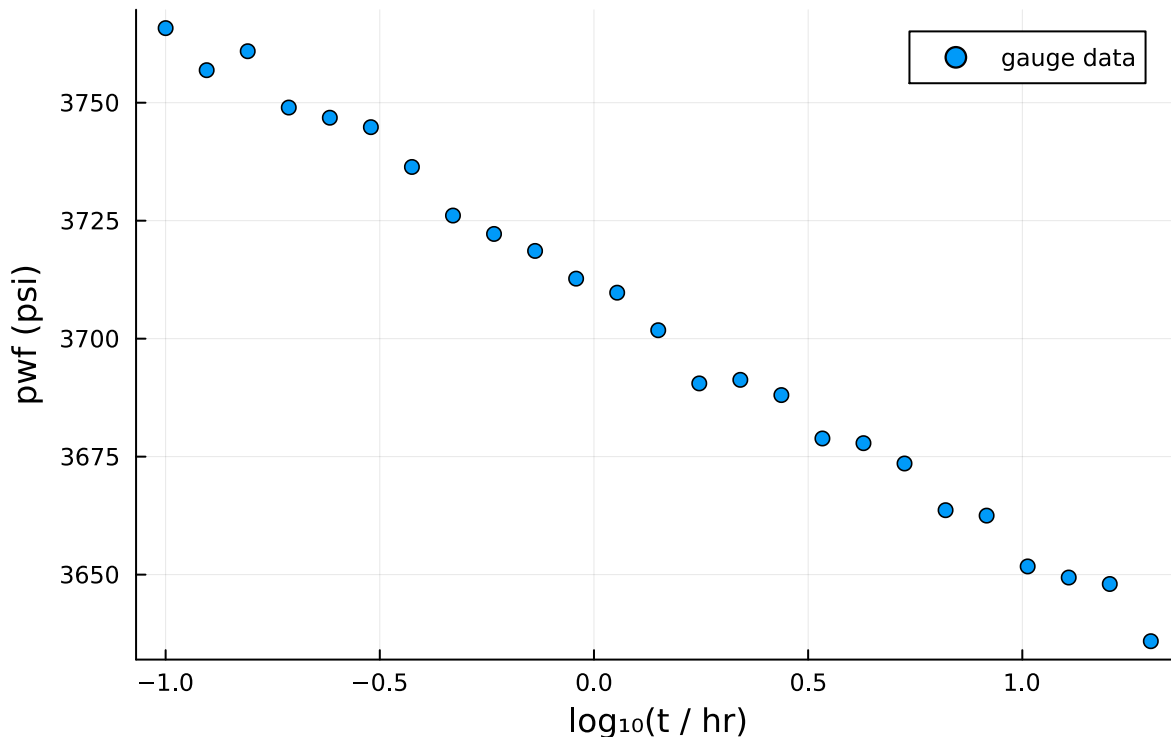
```
1 begin
2   t_demo = 10 .^ range(-1, 1.5; length=100) # hours, log-spaced like a real gauge
3   plot(t_demo, pwf_model.(t_demo, k_demo, s_demo); xscale=:log10, xlabel="t (hr)",
4         ylabel="pwf (psi)",
5         label="k=$(k_demo) md, s=$(s_demo)", title="Simulated drawdown",
6         legend=:bottomleft)
7 end
```

3. Synthetic buildup data ⇄

We generate a noisy gauge record from "true" reservoir properties that we then try to recover blind.

▶ [3765.8, 3756.89, 3760.91, 3748.98, 3746.82, 3744.82, 3736.38, 3726.09, 3722.19, 3718.59, 3712.19, 3708.59, 3704.98, 3701.38, 3697.78, 3694.18, 3690.58, 3686.98, 3683.38, 3679.78, 3676.18, 3672.58, 3668.98, 3665.38, 3661.78, 3658.18, 3654.58, 3650.98, 3647.38, 3643.78, 3640.18, 3636.58, 3632.98, 3629.38, 3625.78, 3622.18, 3618.58, 3614.98, 3611.38, 3607.78, 3604.18, 3600.58, 3596.98, 3593.38, 3589.78, 3586.18, 3582.58, 3578.98, 3575.38, 3571.78, 3568.18, 3564.58, 3560.98, 3557.38, 3553.78, 3550.18, 3546.58, 3542.98, 3539.38, 3535.78, 3532.18, 3528.58, 3524.98, 3521.38, 3517.78, 3514.18, 3510.58, 3506.98, 3503.38, 3499.78, 3496.18, 3492.58, 3488.98, 3485.38, 3481.78, 3478.18, 3474.58, 3470.98, 3467.38, 3463.78, 3460.18, 3456.58, 3452.98, 3449.38, 3445.78, 3442.18, 3438.58, 3434.98, 3431.38, 3427.78, 3424.18, 3420.58, 3416.98, 3413.38, 3409.78, 3406.18, 3402.58, 3398.98, 3395.38, 3391.78, 3388.18, 3384.58, 3380.98, 3377.38, 3373.78, 3370.18, 3366.58, 3362.98, 3359.38, 3355.78, 3352.18, 3348.58, 3344.98, 3341.38, 3337.78, 3334.18, 3330.58, 3326.98, 3323.38, 3319.78, 3316.18, 3312.58, 3308.98, 3305.38, 3301.78, 3298.18, 3294.58, 3290.98, 3287.38, 3283.78, 3280.18, 3276.58, 3272.98, 3269.38, 3265.78, 3262.18, 3258.58, 3254.98, 3251.38, 3247.78, 3244.18, 3240.58, 3236.98, 3233.38, 3229.78, 3226.18, 3222.58, 3218.98, 3215.38, 3211.78, 3208.18, 3204.58, 3200.98, 3197.38, 3193.78, 3190.18, 3186.58, 3182.98, 3179.38, 3175.78, 3172.18, 3168.58, 3164.98, 3161.38, 3157.78, 3154.18, 3150.58, 3146.98, 3143.38, 3139.78, 3136.18, 3132.58, 3128.98, 3125.38, 3121.78, 3118.18, 3114.58, 3110.98, 3107.38, 3103.78, 3099.38, 3095.38, 3091.38, 3087.38, 3083.38, 3079.38, 3075.38, 3071.38, 3067.38, 3063.38, 3059.38, 3055.38, 3051.38, 3047.38, 3043.38, 3039.38, 3035.38, 3031.38, 3027.38, 3023.38, 3019.38, 3015.38, 3011.38, 3007.38, 3003.38, 2999.38, 2995.38, 2991.38, 2987.38, 2983.38, 2979.38, 2975.38, 2971.38, 2967.38, 2963.38, 2959.38, 2955.38, 2951.38, 2947.38, 2943.38, 2939.38, 2935.38, 2931.38, 2927.38, 2923.38, 2919.38, 2915.38, 2911.38, 2907.38, 2903.38, 2899.38, 2895.38, 2891.38, 2887.38, 2883.38, 2879.38, 2875.38, 2871.38, 2867.38, 2863.38, 2859.38, 2855.38, 2851.38, 2847.38, 2843.38, 2839.38, 2835.38, 2831.38, 2827.38, 2823.38, 2819.38, 2815.38, 2811.38, 2807.38, 2803.38, 2799.38, 2795.38, 2791.38, 2787.38, 2783.38, 2779.38, 2775.38, 2771.38, 2767.38, 2763.38, 2759.38, 2755.38, 2751.38, 2747.38, 2743.38, 2739.38, 2735.38, 2731.38, 2727.38, 2723.38, 2719.38, 2715.38, 2711.38, 2707.38, 2703.38, 2699.38, 2695.38, 2691.38, 2687.38, 2683.38, 2679.38, 2675.38, 2671.38, 2667.38, 2663.38, 2659.38, 2655.38, 2651.38, 2647.38, 2643.38, 2639.38, 2635.38, 2631.38, 2627.38, 2623.38, 2619.38, 2615.38, 2611.38, 2607.38, 2603.38, 2599.38, 2595.38, 2591.38, 2587.38, 2583.38, 2579.38, 2575.38, 2571.38, 2567.38, 2563.38, 2559.38, 2555.38, 2551.38, 2547.38, 2543.38, 2539.38, 2535.38, 2531.38, 2527.38, 2523.38, 2519.38, 2515.38, 2511.38, 2507.38, 2503.38, 2499.38, 2495.38, 2491.38, 2487.38, 2483.38, 2479.38, 2475.38, 2471.38, 2467.38, 2463.38, 2459.38, 2455.38, 2451.38, 2447.38, 2443.38, 2439.38, 2435.38, 2431.38, 2427.38, 2423.38, 2419.38, 2415.38, 2411.38, 2407.38, 2403.38, 2399.38, 2395.38, 2391.38, 2387.38, 2383.38, 2379.38, 2375.38, 2371.38, 2367.38, 2363.38, 2359.38, 2355.38, 2351.38, 2347.38, 2343.38, 2339.38, 2335.38, 2331.38, 2327.38, 2323.38, 2319.38, 2315.38, 2311.38, 2307.38, 2303.38, 2299.38, 2295.38, 2291.38, 2287.38, 2283.38, 2279.38, 2275.38, 2271.38, 2267.38, 2263.38, 2259.38, 2255.38, 2251.38, 2247.38, 2243.38, 2239.38, 2235.38, 2231.38, 2227.38, 2223.38, 2219.38, 2215.38, 2211.38, 2207.38, 2203.38, 2199.38, 2195.38, 2191.38, 2187.38, 2183.38, 2179.38, 2175.38, 2171.38, 2167.38, 2163.38, 2159.38, 2155.38, 2151.38, 2147.38, 2143.38, 2139.38, 2135.38, 2131.38, 2127.38, 2123.38, 2119.38, 2115.38, 2111.38, 2107.38, 2103.38, 2099.38, 2095.38, 2091.38, 2087.38, 2083.38, 2079.38, 2075.38, 2071.38, 2067.38, 2063.38, 2059.38, 2055.38, 2051.38, 2047.38, 2043.38, 2039.38, 2035.38, 2031.38, 2027.38, 2023.38, 2019.38, 2015.38, 2011.38, 2007.38, 2003.38, 1999.38, 1995.38, 1991.38, 1987.38, 1983.38, 1979.38, 1975.38, 1971.38, 1967.38, 1963.38, 1959.38, 1955.38, 1951.38, 1947.38, 1943.38, 1939.38, 1935.38, 1931.38, 1927.38, 1923.38, 1919.38, 1915.38, 1911.38, 1907.38, 1903.38, 1899.38, 1895.38, 1891.38, 1887.38, 1883.38, 1879.38, 1875.38, 1871.38, 1867.38, 1863.38, 1859.38, 1855.38, 1851.38, 1847.38, 1843.38, 1839.38, 1835.38, 1831.38, 1827.38, 1823.38, 1819.38, 1815.38, 1811.38, 1807.38, 1803.38, 1799.38, 1795.38, 1791.38, 1787.38, 1783.38, 1779.38, 1775.38, 1771.38, 1767.38, 1763.38, 1759.38, 1755.38, 1751.38, 1747.38, 1743.38, 1739.38, 1735.38, 1731.38, 1727.38, 1723.38, 1719.38, 1715.38, 1711.38, 1707.38, 1703.38, 1699.38, 1695.38, 1691.38, 1687.38, 1683.38, 1679.38, 1675.38, 1671.38, 1667.38, 1663.38, 1659.38, 1655.38, 1651.38, 1647.38, 1643.38, 1639.38, 1635.38, 1631.38, 1627.38, 1623.38, 1619.38, 1615.38, 1611.38, 1607.38, 1603.38, 1599.38, 1595.38, 1591.38, 1587.38, 1583.38, 1579.38, 1575.38, 1571.38, 1567.38, 1563.38, 1559.38, 1555.38, 1551.38, 1547.38, 1543.38, 1539.38, 1535.38, 1531.38, 1527.38, 1523.38, 1519.38, 1515.38, 1511.38, 1507.38, 1503.38, 1499.38, 1495.38, 1491.38, 1487.38, 1483.38, 1479.38, 1475.38, 1471.38, 1467.38, 1463.38, 1459.38, 1455.38, 1451.38, 1447.38, 1443.38, 1439.38, 1435.38, 1431.38, 1427.38, 1423.38, 1419.38, 1415.38, 1411.38, 1407.38, 1403.38, 1399.38, 1395.38, 1391.38, 1387.38, 1383.38, 1379.38, 1375.38, 1371.38, 1367.38, 1363.38, 1359.38, 1355.38, 1351.38, 1347.38, 1343.38, 1339.38, 1335.38, 1331.38, 1327.38, 1323.38, 1319.38, 1315.38, 1311.38, 1307.38, 1303.38, 1299.38, 1295.38, 1291.38, 1287.38, 1283.38, 1279.38, 1275.38, 1271.38, 1267.38, 1263.38, 1259.38, 1255.38, 1251.38, 1247.38, 1243.38, 1239.38, 1235.38, 1231.38, 1227.38, 1223.38, 1219.38, 1215.38, 1211.38, 1207.38, 1203.38, 1199.38, 1195.38, 1191.38, 1187.38, 1183.38, 1179.38, 1175.38, 1171.38, 1167.38, 1163.38, 1159.38, 1155.38, 1151.38, 1147.38, 1143.38, 1139.38, 1135.38, 1131.38, 1127.38, 1123.38, 1119.38, 1115.38, 1111.38, 1107.38, 1103.38, 1099.38, 1095.38, 1091.38, 1087.38, 1083.38, 1079.38, 1075.38, 1071.38, 1067.38, 1063.38, 1059.38, 1055.38, 1051.38, 1047.38, 1043.38, 1039.38, 1035.38, 1031.38, 1027.38, 1023.38, 1019.38, 1015.38, 1011.38, 1007.38, 1003.38, 999.38, 995.38, 991.38, 987.38, 983.38, 979.38, 975.38, 971.38, 967.38, 963.38, 959.38, 955.38, 951.38, 947.38, 943.38, 939.38, 935.38, 931.38, 927.38, 923.38, 919.38, 915.38, 911.38, 907.38, 903.38, 899.38, 895.38, 891.38, 887.38, 883.38, 879.38, 875.38, 871.38, 867.38, 863.38, 859.38, 855.38, 851.38, 847.38, 843.38, 839.38, 835.38, 831.38, 827.38, 823.38, 819.38, 815.38, 811.38, 807.38, 803.38, 799.38, 795.38, 791.38, 787.38, 783.38, 779.38, 775.38, 771.38, 767.38, 763.38, 759.38, 755.38, 751.38, 747.38, 743.38, 739.38, 735.38, 731.38, 727.38, 723.38, 719.38, 715.38, 711.38, 707.38, 703.38, 699.38, 695.38, 691.38, 687.38, 683.38, 679.38, 675.38, 671.38, 667.38, 663.38, 659.38, 655.38, 651.38, 647.38, 643.38, 639.38, 635.38, 631.38, 627.38, 623.38, 619.38, 615.38, 611.38, 607.38, 603.38, 599.38, 595.38, 591.38, 587.38, 583.38, 579.38, 575.38, 571.38, 567.38, 563.38, 559.38, 555.38, 551.38, 547.38, 543.38, 539.38, 535.38, 531.38, 527.38, 523.38, 519.38, 515.38, 511.38, 507.38, 503.38, 499.38, 495.38, 491.38, 487.38, 483.38, 479.38, 475.38, 471.38, 467.38, 463.38, 459.38, 455.38, 451.38, 447.38, 443.38, 439.38, 435.38, 431.38, 427.38, 423.38, 419.38, 415.38, 411.38, 407.38, 403.38, 399.38, 395.38, 391.38, 387.38, 383.38, 379.38, 375.38, 371.38, 367.38, 363.38, 359.38, 355.38, 351.38, 347.38, 343.38, 339.38, 335.38, 331.38, 327.38, 323.38, 319.38, 315.38, 311.38, 307.38, 303.38, 299.38, 295.38, 291.38, 287.38, 283.38, 279.38, 275.38, 271.38, 267.38, 263.38, 259.38, 255.38, 251.38, 247.38, 243.38, 239.38, 235.38, 231.38, 227.38, 223.38, 219.38, 215.38, 211.38, 207.38, 203.38, 199.38, 195.38, 191.38, 187.38, 183.38, 179.38, 175.38, 171.38, 167.38, 163.38, 159.38, 155.38, 151.38, 147.38, 143.38, 139.38, 135.38, 131.38, 127.38, 123.38, 119.38, 115.38, 111.38, 107.38, 103.38, 99.38, 95.38, 91.38, 87.38, 83.38, 79.38, 75.38, 71.38, 67.38, 63.38, 59.38, 55.38, 51.38, 47.38, 43.38, 39.38, 35.38, 31.38, 27.38, 23.38, 19.38, 15.38, 11.38, 7.38, 3.38, -1.38, -5.38, -9.38, -13.38, -17.38, -21.38, -25.38, -29.38, -33.38, -37.38, -41.38, -45.38, -49.38, -53.38, -57.38, -61.38, -65.38, -69.38, -73.38, -77.38, -81.38, -85.38, -89.38, -93.38, -97.38, -101.38, -105.38, -109.38, -113.38, -117.38, -121.38, -125.38, -129.38, -133.38, -137.38, -141.38, -145.38, -149.38, -153.38, -157.38, -161.38, -165.38, -169.38, -173.38, -177.38, -181.38, -185.38, -189.38, -193.38, -197.38, -201.38, -205.38, -209.38, -213.38, -217.38, -221.38, -225.38, -229.38, -233.38, -237.38, -241.38, -245.38, -249.38, -253.38, -257.38, -261.38, -265.38, -269.38, -273.38, -277.38, -281.38, -285.38, -289.38, -293.38, -297.38, -301.38, -305.38, -309.38, -313.38, -317.38, -321.38, -325.38, -329.38, -333.38, -337.38, -341.38, -345.38, -349.38, -353.38, -357.38, -361.38, -365.38, -369.38, -373.38, -377.38, -381.38, -385.38, -389.38, -393.38, -397.38, -401.38, -405.38, -409.38, -413.38, -417.38, -421.38, -425.38, -429.38, -433.38, -437.38, -441.38, -445.38, -449.38, -453.38, -457.38, -461.38, -465.38, -469.38, -473.38, -477.38, -481.38, -485.38, -489.38, -493.38, -497.38, -501.38, -505.38, -509.38, -513.38, -517.38, -521.38, -525.38, -529.38, -533.38, -537.38, -541.38, -545.38, -549.38, -553.38, -557.38, -561.38, -565.38, -569.38, -573.38, -577.38, -581.38, -585.38, -589.38, -593.38, -597.38, -601.38, -605.38, -609.38, -613.38, -617.38, -621.38, -625.38, -629.38, -633.38, -637.38, -641.38, -645.38, -649.38, -653.38, -657.38, -661.38, -665.38, -669.38, -673.38, -677.38, -681.38, -685.38, -689.38, -693.38, -697.38, -701.38, -705.38, -709.38, -713.38, -717.38, -721.38, -725.38, -729.38, -733.38, -737.38, -741.38, -745.38, -749.38, -753.38, -757.38, -761.38, -765.38, -769.38, -773.38, -777.38, -781.38, -785.38, -789.38, -793.38, -797.38, -801.38, -805.38, -809.38, -813.38, -817.38, -821.38, -825.38, -829.38, -833.38, -837.38, -841.38, -845.38, -849.38, -853.38, -857.38, -861.38, -865.38, -869.38, -873.38, -877.38, -881.38, -885.38, -889.38, -893.38, -897.38, -901.38, -905.38, -909.38, -913.38, -917.38, -921.38, -925.38, -929.38, -933.38, -937.38, -941.38, -945.38, -949.38, -953.38, -957.38, -961.38, -965.38, -969.38, -973.38, -977.38, -981.38, -985.38, -989.38, -993.38, -997.38, -1001.38, -1005.38, -1009.38, -1013.38, -1017.38, -1021.38, -1025.38, -1029.38, -1033.38, -1037.38, -1041.38, -1045.38, -1049.38, -1053.38, -1057.38, -1061.38, -1065.38, -1069.38, -1073.38, -1077.38, -1081.38, -1085.38, -1089.38, -1093.38, -1097.38, -1101.38, -1105.38, -1109.38, -1113.38, -1117.38, -1121.38, -1125.38, -1129.38, -1133.38, -1137.38, -1141.38, -1145.38, -1149.38, -1153.38, -1157.38, -1161.38, -1165.38, -1169.38, -1173.38, -1177.38, -1181.38, -1185.38, -1189.38, -1193.38, -1197.38, -1201.38, -1205.38, -1209.38, -1213.38, -1217.38, -1221.38, -1225.38, -1229.38, -1233.38, -1237.38, -1241.38, -1245.38, -1249.38, -1253.38, -1257.38, -1261.38, -1265.38, -1269.38, -1273.38, -1277.38, -1281.38, -1285.38, -1289.38, -1293.38, -1297.38, -1301.38, -1305.38, -1309.38, -1313.38, -1317.38, -1321.38, -1325.38, -1329.38, -1333.38, -1337.38, -1341.38, -1345.38, -1349.38, -1353.38, -1357.38, -1361.38, -1365.38, -1369.38, -1373.38, -1377.38, -1381.38, -1385.38, -1389.38, -1393.38, -1397.38, -1401.38, -1405.38, -1409.38, -1413.38, -1417.38, -1421.38, -1425.38, -1429.38, -1433.38, -1437.38, -1441.38, -1445.38, -1449.38, -1453.38, -1457.38, -1461.38, -1465.38, -1469.38, -1473.38, -1477.38, -1481.38, -1485.38, -1489.38, -1493.38, -1497.38, -1501.38, -1505.38, -1509.38, -1513.38, -1517.38, -1521.38, -1525.38, -1529.38, -1533.38, -1537.38, -1541.38, -1545.38, -1549.38, -1553.38, -1557.38, -1561.38, -1565.38, -1569.38, -1573.38, -1577.38, -1581.38, -1585.38, -1589.38, -1593.38, -1597.38, -1601.38, -1605.38, -1609.38, -1613.38, -1617.38, -1621.38, -1625.38, -1629.38, -1633.38, -1637.38, -1641.38, -1645.38, -1649.38, -1653.38, -1657.38, -1661.38, -1665.38, -1669.38

Semi-log plot (the classic MDH straight line)



```
1 scatter(log10.(t_test), pwf_observed; label="gauge data", xlabel="log10(t / hr)",
  ylabel="pwf (psi)",
2 title="Semi-log plot (the classic MDH straight line)")
```

4. Estimate k and s by nonlinear regression ⇔

This is exactly the semi-log straight-line method, just fit by least squares instead of a ruler: the model is linear in $\ln t$, so the fit is extremely well-conditioned.

welltest_loss (generic function with 1 method)

```
1 function welltest_loss(logp, _)
2     k = exp(logp[1])
3     s = logp[2]
4     pred = pwf_model.(t_test, k, s)
5     return sum(abs2, pwf_observed .- pred)
6 end
```

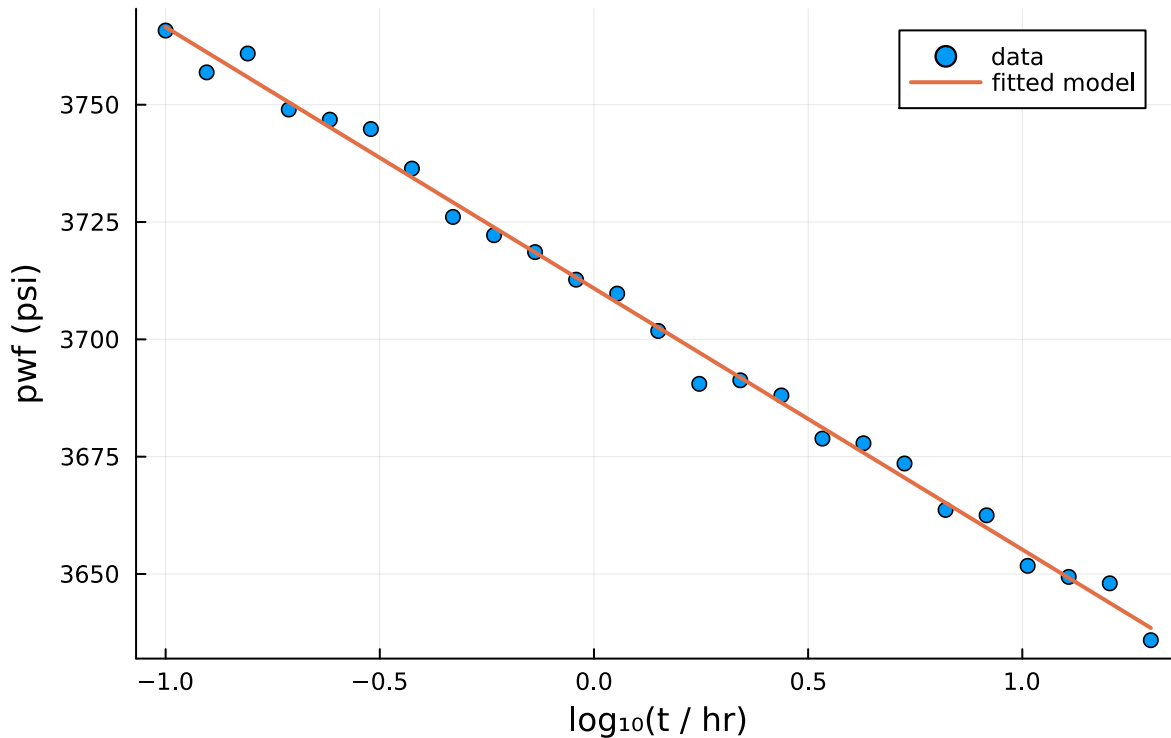
► SciMLBase.OptimizationSolution{Float64, 1, Vector{Float64}, OptimizationBase.OptimizationCache{Op

```
1 begin
2     p0_welltest = [log(10.0), 0.0]
3     optf_welltest = OptimizationFunction(welltest_loss, Optimization.AutoForwardDiff())
4     optprob_welltest = OptimizationProblem(optf_welltest, p0_welltest)
5     fit_welltest = Optimization.solve(optprob_welltest, OptimizationOptimJL.LBFGS())
6 end
```

```
► (fitted_k = 35.0529, fitted_s = 4.52548, true_k = 35.0, true_s = 4.5)
```

```
1 (fitted_k=exp(fit_welltest.u[1]), fitted_s=fit_welltest.u[2], true_k=true_k,  
   true_s=true_s)
```

Recovered k = 35.1 md, s = 4.53



```
1 begin  
2   scatter(log10.(t_test), pwf_observed; label="data", xlabel="log10(t / hr)",  
            ylabel="pwf (psi)")  
3   fitted_k_1 = exp(fit_welltest.u[1])  
4   plot!(log10.(t_test), pwf_model.(t_test, fitted_k_1, fit_welltest.u[2]);  
          label="fitted model", lw=2,  
          title="Recovered k = $(round(fitted_k_1, digits=1)) md, s =  
                $(round(fit_welltest.u[2], digits=2))")  
6 end
```

Takeaways ⇨

- The semi-log straight-line method used throughout well-test analysis is nothing more than fitting a known closed-form model to data — Optimization.jl does it without needing to identify the straight-line region by eye.
- Because the model here is an explicit formula rather than an ODE solve, no sensitivity-analysis machinery is needed at all — plain ForwardDiff through the formula is enough.
- The same nonlinear-regression pattern extends directly to more complex well-test models (dual-porosity, finite-conductivity fractures, bounded reservoirs) — only pwf_model needs to change; the fitting code stays identical.

